2

The Biology of Use and Disuse

Statements about use have histories of use. "Form follows function" is such a statement. It is a statement of use that is so well used that it can be hard to think of it as having its own history, its own story of coming into use. Of course, it has such a history. The statement was first used by the architect Louis Sullivan in 1896. We might assume that "form follows function" implies a normative commitment as a principle of design: that form *should* follow function; how a building is built *should* be determined by the function of that building. However, in making this statement, Sullivan was in fact describing what he called simply "the law," that in life, form follows function; indeed, form *ever* follows function.

> Whether it be the sweeping eagle in his flight or the open apple-blossom, the toiling work-horse, the blithe swan, the branching oak, the winding stream at its base, the drifting clouds, over all the coursing sun, form ever follows function, and this is the law. . . . It is the pervading law of all things organic and inorganic, of all things physical and metaphysical, of all things human and all things superhuman, of all true manifestations of the head, of the heart, of the soul, that life is recognizable in its expression, that form ever follows function. This is the law. (1896, 408)

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The idea that form follows function is a natural law is a commitment to a principle: that use and life are entangled. Use is treated as a generative or life principle, that is, as a principle that shapes the very forms of life. We might describe such forms of life as biology. The word *biology* is used for the study for living organisms, including their structure, function, growth, evolution, distribution, identification, and taxonomy. As Sarah Franklin has noted, biology can refer to both a "body of authoritative knowledge (as in the science of reproductive biology) and a set of phenomena" (2001, 303). Biology can thus refer both to studies of living organisms and to the living organisms themselves. It is noteworthy how early studies of living organisms made use of use to explain a set of phenomena. Jean-Baptiste Lamarck, the first scholar to use the word *biology* in a scientific sense, describes the laws of use and disuse as "laws of nature which are always verified by observation" ([1809] 1914, 113).[1] This primacy is pedagogy: biology and use coevolved right from the beginning.

In this chapter, I compare how Lamarck and Darwin made use of use in their explanation of the origin of species. Why this comparison? I noted in my introduction to this book that *use* is a much-used word: if we follow the word, we would end up all over the place. But when use is used in order to explain phenomena, use is being channeled in specific directions. Comparing Lamarck and Darwin's uses of use provides a way of *zooming in*, allowing me to explore what use is doing when exercised as a conceptual tool. A comparison between Lamarck and Darwin provides a particularly effective case study of the uses of use because the differences and similarities between them have already been framed in these terms. For example, Ernst Mayr counted the number of times Darwin evokes "use and disuse" in *The Origin of Species* as a way of demonstrating the extent of Lamarck's influence on Darwin (Mayr 1972, 78).[2] That the relation between two scholars can be decided by counting the number of times one made use of the other's law of use and disuse has much to teach us. In this chapter, I consider how Lamarck and Darwin mobilized the vocabulary of use before discussing how the figure of the blacksmith's arm was used to exemplify Lamarckian principles even though Lamarck himself did not use this example. The figure of the blacksmith's arm provides a handy way of showing how and why biological and sociological thought in the early nineteenth century invested so much significance in use not only as a generative or life principle but as a method of social improvement and advancement.

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The Law of Use and Disuse

In *Zoological Philosophy*, Lamarck makes a case for a functionalist approach to biology rather than what would typically be characterized as a structuralist approach. He argues, "Naturalists have remarked that the structure of animals is always in perfect adaptation to their functions, and have inferred that the shape and condition of parts have determined their use of them" ([1809] 1914, 113). Lamarck argues against any such inference. He suggests from observation that "the needs and uses of the parts that has caused the development of these same parts, which have even given birth to them when they did not exist, and which have consequently given rise to the development of these same parts" (113). A need and use for a part not only shapes a part but can bring it into existence. I call this principle as it is articulated in relation to designed objects "for is before."

What Lamarck presents first as an observation provides the basis of a law, what he calls the law of use and disuse: "*In every animal which has not passed the limit of its development, a more frequent and continuous use of any organ gradually strengthens, develops and enlarges that organ, and gives it a power proportional to the length of time it has been so used; while the permanent disuse of any organ imperceptibly weakens and deteriorates it, and progressively diminishes its functional capacity, until it finally disappears*" (113). The laws condense some of the arguments discussed in chapter 1 about how using something strengthens it (here "something" refers to an organ), and how the disuse of something can lead to its weakening and eventual disappearance. Lamarck's thesis is explicitly framed in terms of time; use refers to the length of time being used. And time is related to power: the more time something is used, the more power it acquires; the less time something is used, the less power it retains. Power is time in proportion. To make use of a part is to take time: use as strengthening and preserving; disuse as weakening and withering.

It is important to note that Lamarck's first law about the effects of use and disuse was not introduced by Lamarck. He himself observed that "the law of the effects of exercise on life has been grasped for a long time by observers attentive to the phenomena of organization" (cited in Burkhardt [1977] 1995, 166). The law of exercise simply suggests that the more something is done, the better it is done, or the more an association is made, the stronger it becomes. Note here how the law of exercise is referred to as central to the "phenomena of organisation," such that use and disuse can be understood as organizational principles.

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Lamarck's law of use and disuse is presented as empirical observation: what can be grasped from attention to the organization of life. Use and disuse are also used to explain how individual organisms gradually come to take their shape over long periods of time; such shaping by use and disuse is not easily perceptible from the vantage point of the present. The second law is that of use inheritance. Simply put, Lamarck argues that the effects of use and disuse are inherited by future generations if certain reproductive conditions are met: "*All the acquisitions or losses wrought by nature on individuals, through the influence of the environment in which their race has long been placed, and hence through the influence of the predominant use or permanent disuse of any organ; all these are preserved by reproduction to the new individuals which arise, provided that the acquired modifications are common to both sexes, or at least to the individuals which produce the young*" ([1809] 1914, 113). Use inheritance is the inheritance of form acquired as a result of repeated use. Use and disuse are themselves understood as *acquired modifications* that can be transmitted over time, which is how use and disuse come to shape not only individual being but species being.

It is important to note that for Lamarck, use and disuse are *effects*, however much they affect the structure of organisms. "Form follows function" thus describes only one aspect of Lamarck's thesis; even if function is given precedence over form, function comes after or even follows the environment. By "following" here we would be concerned with *influence*. Lamarck describes the influence of the environment on the habits and activities of animals, and the influence of habits and activities of animals on their organization and structure. Use and disuse are thus framed as "habits and activities." I have described use as the building block of habit, the "small steps" that take us in a certain direction. Habit is defined by Lamarck as "*second nature*" ([1809] 1914, 114). When use is repeated, becoming a pattern, use becomes second nature.

For Lamarck, habit is shaped by habitat. And habitats are dynamic. This means that for Lamarck, use and disuse are not first principles; that is, they do not start the endless process of modification, which is evident from the letter of the law: "*hence* through the influence of the predominant use or permanent disuse of any organ" ([1809] 1914, 113, emphasis mine). Use and disuse are how animals are, as it were, under the influence; they could be described as how organisms receive a message from the environment in the form of needs. It is important to Lamarck's argument that the environment's effect on structure is not direct: "Whatever the environment may do it does not work any direct modification

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whatever in the shape and organisation of animals" (107). In other words, he is not suggesting that the environment causes in an efficient manner changes to biological structure.[3]

I would suggest that Lamarck makes use of use in order to make sense of how modifications are achieved *indirectly*. Even if the environment does not directly modify the structure of organisms, the environment is decisive because it shapes what an organism *needs* in order to survive. Lamarck writes, "New needs which establish the necessity for some part really bring about the existence of that part" ([1809] 1914, 108). I do not think there is a claim here that a need for something is an efficient cause; it might be more useful to understand the claim with reference to what Aristotle called a final cause.[4] Although Lamarck is primarily discussing use as activity or employment (*emploi*), his emphasis on needs (*besoin*) could also be understood as referring to use in another sense.[5] The English word *use* can be used as an idiom to refer to need (to have a use for something as a need of something), which means in English we can use the same word (*use*) to refer to need and to employment.[6] By emphasizing need, Lamarck is making another use of use. Simply put: for Lamarck, a part is used more *because* there is more of a use for it; use as employment follows use as need.

So already we have two levels in which use is understood as generative: use as an activity that modifies an organism (a modification that can be inherited) and use as directing activities in this way or that (use as a requirement). It is the environment that makes certain activities necessary for survival, which is to say, which determines what would be more or less useful for organisms to do and to have. If the use and disuse of parts shape the form of organisms, use and disuse are shaped by what would be useful or not useful *given* an environment. It is given this given that use and disuse are themselves not given, in advance, as it were.

Use in both the senses of need and employment is thus an effect more than a cause, or an effect that becomes a cause: an emergence might be triggered by a state of emergency (for example, by a change from an existing habitat or migration to a different habitat) in which something that is necessary does not yet exist. *Use is a way of showing how what is necessary is contingent.* Simply put: it is because environments are dynamic and changing that organisms need to be dynamic and changing. Indeed, the emphasis in his description is how environmental change requires alterations in activities: "Great alterations in the environment of animals lead to great alterations in their needs, and these alterations in their needs

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necessarily lead to others in their activities" ([1809] 1914, 107). The environment is put in the position of the subject who is doing things (in an earlier sentence, Lamarck describes how "*it* produces in course of time corresponding modification in the shape and organisation of animals"). Once a new habit is formed, habit shapes form: "If a new environment, which has become permanent for some race of animals, induces new habits in these animals, that is to say, leads them to new activities which have become habitual, the result will be the use of some one part in preference to some other part, and in some cases the total disuse of some part no longer necessary" (108). For Lamarck, use is how a body receives *and* how it acts: a body receives the influence of an environment in the requirement to act in a certain way, to make use of some parts more than others. This "making use" is transformative.

One of Lamarck's most famous examples is that of the giraffe's long neck. I return to the status of this example in the third section of this chapter. Lamarck suggests that the giraffe's neck becomes longer over time as an effect of the sustained effort to reach the higher foliage that the giraffe needs to reach (122). This is not a question of willing what is necessary.[7] It is question of the *effects* of doing what is necessary to survive. Use is both how an organism receives a message from the environment about what it needs as well as how an organism is directed to act in particular ways. We can bring these arguments together: *reception is direction.* Indeed, lengthening would be understood as how effort shapes form in the sense of giving direction: "Efforts in a given direction, when they are long sustained or habitually made by certain parts of a living body, for the satisfaction of needs established by nature or environment, cause an enlargement of the parts and the acquisition of a size and shape that they would never have obtained, *if these efforts had not become the normal activities of the animal exerting them*" (Lamarck [1809] 1914, 123, emphasis mine). When an effort becomes normal, a form is acquired. When a form is acquired, less effort is needed.

In the previous chapter, I contrasted a used and an unused path, suggesting that the use can smooth and ease a passage. When a path is used more, it is easier to follow. The well-used path provides a pictorial snapshot of the *longue durée* described by Lamarck. Behavior can be understood as a path, a route that is being followed. The incentive to create new paths, *to do what does not come naturally*, derives from the pressures of environmental change. Gradually, over time, a path becomes easier, which is to say, the effort required is lessened. This lessening *is* the acquisition

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2.1. A longer neck.

of form. A form such as a longer neck is a well-used path, *an effect of use that eases use* (figure 2.1).

The implication of Lamarck's argument is that changes in behavior triggered by environmental changes precede changes in form. Stephen Jay Gould argued that it was this aspect of Lamarck's thesis that should be understood as his "primary contribution" (1985, 36). Gould also suggests that the lag between behavioral change and formal change accounts for some of the "oddest" and most "curious" of animal inventions. Gould's examples include flamingos that dwell in hypersaline lakes: "Few creatures can tolerate the unusual environments of these saline deserts" (24). The flamingos have an unusual way of feeding "with their heads upside down" (25). This "flip flop" is described as "complete and comprehensive" not only in form but also in motion (32). The action is also described as "topsy-turvy" (32). Gould concludes that a "peculiar reversal in behavior has engendered a complex inversion of form" (32).

Such curious reversals and complex inversions offer another lens with which to explore the queerness of biology. Scholars not only have explored queer biology by finding evidence of homosexual pairings in the animal world or by pointing to organisms that can change sex or to nonsexual

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forms of reproduction but have also shown how these forms of diversity can attune us to other forms of social complexity and creativity (Bagemihl 1999; Roughgarden 2004).[8] We can follow their lead by using queer biology to refer to what Gould calls a "peculiar reversal in behavior" and a "complex inversion of form." The word *inversion* was used in the nineteenth century to describe homosexual attachments: in the language of sexology, the inverts felt internally that they were the "opposite sex." The word *invert* was used because it implied being *turned* in the wrong direction. We can pick up on the old meanings of that word, just as we can pick up on the old meanings of queer. An inversion can teach us about how the right direction is achieved or how the upright becomes the right way up.[9] In order to survive in an unusual environment, the flamingo's beak ends up the wrong way up. An implication of such curious cases might be that usual orientations or even a sense of there being a "right way up" are effects of habits; straight or upright become second nature in time.

Something becomes queer, odd, noticeable, or curious when it reverses how things exist usually or by challenging how things are expected to be. When something has become usual, the unusual is striking. It is not surprising that the queerness of biology is brought out by attending to uses of use. After all, adaptation could itself be described as a process of fitting (*adapt* comes from Latin, *adaptāre*, to fit or adjust), which means *not fitting* acquires originary status (in the sense that not fitting is what requires adaption). In chapter 1, I discussed how not fitting is understood as an incentive to design new objects. In biology, not fitting is given a similar role as "an impetus to change," to borrow Gould's terms. He elaborates, "An organism enters a new environment with its old form suited to other styles of life. The behavioral innovation establishes a discordance between new function and inherited form—an impetus to change" (1985, 36–37). A discord is temporal, between past and present as well as being a discord between form and function. If organisms and environments are always changing, they are not changing at the same time. *The nonsimultaneity of change is how biology and time become queer at the same time.* The queerness of biology, the existence of curious reversals and inverted forms, can be exposed through the queering of time, the nonaccordance of past and present. We can draw on Elizabeth Freeman's (2010) work on queer temporality as "temporal drag." Forms could be thought of as "temporal drag"; the visceral pull of a past is evidenced by how they drag behind functions, a dragging that is expressed queerly in "imperfections and odd solutions cobbled together from parts on hand" (Gould 1985, 37).

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If norms *become* forms in time, then norms are not *always* forms. Organisms become more attuned to their environment over time, which means nonattunement—not fitting the requirements—is a necessary part of a struggle for existence. I noted in chapter 1 how attunement can reduce the effort required to complete a task. Attunement can also mean the reduction of the effort to do what is necessary to survive. The implication of use inheritance is that a reduction of effort is transmitted *as form*, which means that organisms, by inheriting a history, are *inheriting an effort*. I return in chapter 4 to the implications of this argument for an understanding of social inheritance.

Usefulness and Selection

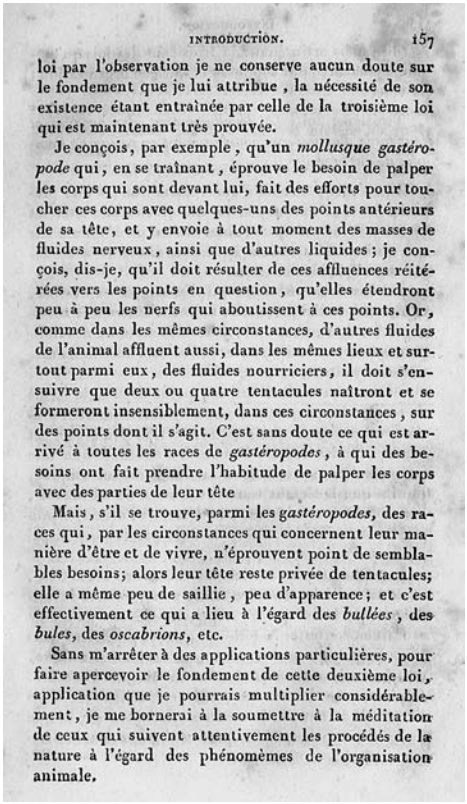
Lamarck certainly implied that use can be a reason for existence: a use for something can bring it about. Charles Darwin seemed to deviate from Lamarck because of the implication he heard that organisms can bring about what they need.[10] Richard Burkhardt's *The Spirit of System* refers to the following sentence by Lamarck: "The forces brought into being by a newly felt need will necessarily give rise to the proper organ for satisfying this new need if the organ does not exist already." Burkhardt notes, "In his copy of the second edition of Lamarck's *Histoire naturelle*, Charles Darwin responded to Lamarck's claim by jotting on the page where the claim appeared: 'Because use improves an organ, wishing for it, or its use, produces it!!! oh'" ([1977] 1995, 178).

It is interesting how Darwin parts company from Lamarck on the specific question of use. And we can pause here with the punctuation, that is, with Darwin's own use of three exclamation points.

!!!

In the previous chapter, I explored how exclamation points are used in order to give emphasis, or to give more intensity to an expression. When exclamations are overused, they might cease to function as they are supposed to. One could consider how a history of use can be expressed as a series of exclamation points. That Darwin makes use of three exclamation points has something to teach us about how he judged the book he was reading. The jot is available to read by virtue of digital reproduction; we cannot see the inscription (it is too faint to be reproduced), but we can read the translation (figures 2.2, 2.3).

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INTRODUCTION. 157

loi par l'observation je ne conserve aucun doute sur le fondement que je lui attribue, la nécessité de son existence étant entraînée par celle de la troisième loi qui est maintenant très prouvée.

Je conçois, par exemple, qu'un *mollusque gastéropode* qui, en se traînant, éprouve le besoin de palper les corps qui sont devant lui, fait des efforts pour toucher ces corps avec quelques-uns des points antérieurs de sa tête, et y envoie à tout moment des masses de fluides nerveux, ainsi que d'autres liquides; je conçois, dis-je, qu'il doit résulter de ces affluences réitérées vers les points en question, qu'elles étendront peu à peu les nerfs qui aboutissent à ces points. Or, comme dans les mêmes circonstances, d'autres fluides de l'animal affluent aussi, dans les mêmes lieux et surtout parmi eux, des fluides nourriciers, il doit s'ensuivre que deux ou quatre tentacules naîtront et se formeront insensiblement, dans ces circonstances, sur des points dont il s'agit. C'est sans doute ce qui est arrivé à toutes les races de *gastéropodes*, à qui des besoins ont fait prendre l'habitude de palper les corps avec des parties de leur tête

Mais, s'il se trouve, parmi les *gastéropodes*, des races qui, par les circonstances qui concernent leur manière d'être et de vivre, n'éprouvent point de semblables besoins; alors leur tête reste privée de tentacules; elle a même peu de saillie, peu d'apparence; et c'est effectivement ce qui a lieu à l'égard des *bullées*, des *bules*, des *oscabrions*, etc.

Sans m'arrêter à des applications particulières, pour faire apercevoir le fondement de cette deuxième loi, application que je pourrais multiplier considérablement, je me bornerai à la soumettre à la méditation de ceux qui suivent attentivement les procédés de la nature à l'égard des phénomèmes de l'organisation animale.

2.2. Darwin's copy of Lamarck's *Historie Naturelle*, 157. Courtesy of Biodiversity Heritage Library.



2.3. A disagreement with Lamarck's use of use. Courtesy of Biodiversity Heritage Library.

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One thinks of the erasure of use as part of the history of use: the faint marks that might gradually disappear; traces left behind by a user whose own value means the trace is of value. Digital reproduction can allow us to keep track of what otherwise might have faded from memory as well as from view. We might think again of used books and how they matter as recorders of a history. A message scrawled in the margin by Darwin, which was a scrawling *about* use as well as a sign *of* use, teaches us about how Darwin distanced himself not only from Lamarck but from Lamarck's use of use. A used book, as I explored in chapter 1, can tell us many stories. The traces left behind of Darwin's reading of Lamarck, the rather scornful utterance that shows how he read him, offers another trail in our short history of use.

A widely held (but not universally agreed) view that Darwin became progressively more Lamarckian over his career was in part due to the increasing use of use in the revised editions of *The Origin of Species*, as well as in the stronger claims made in *The Descent of Man*, which not only made use of Lamarck's first law but also the second; he stated explicitly that the effects of use and disuse are "probably inherited."[11] We can consider Darwin's own use of use in the first instance by considering how he employed Lamarck's law in relation to his thesis of natural selection. Take the following quote:

> We should keep in mind, as I have before insisted, that the inherited effects of the increased use of parts, and perhaps of their disuse, will be strengthened by natural selection. For all spontaneous variations in the right direction will thus be preserved; as will those individuals which inherit in the highest degree the effects of the increased and beneficial use of any part. How much to attribute in each particular case to the effects of use, and how much to natural selection, it seems impossible to decide. . . . I may give another instance of a structure which apparently owes its origin exclusively to use or habit. The extremity of the tail in some American monkeys has been converted into a wonderfully perfect prehensile organ, and serves as a fifth hand. ([1859] 2009, 188)

In this combination of sentences, Darwin offers the law of use and disuse and natural selection as two differing explanations of biological form. One is not presented as true, and the other false; rather, each is offered as a limited or partial explanation of more or less the same phenomena. Darwin notes, "When a variation is of the slightest use to any

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being, *we cannot tell* how much to attribute to the accumulative action of natural selection, and how much to the definite action of the conditions of life" ([1859] 2009, 107, emphasis mine). This inability to tell how much a variation that is of use is due to natural selection is telling. In both quotes, attribute is not a noun but a verb: some things (variations) can be attributed more or less to conditions of life, which include the effects of use and disuse, or natural selection. What is interesting here is the implied *combination* of effects, as a combination that will cohere differently in different parts. The effects of use and disuse are "strengthened" by natural selection because spontaneous variations in the right direction will be preserved. In his later book *The Descent of Man*, Darwin describes natural selection as "aided" by the effects of use and disuse: "Natural selection would probably have been greatly aided by the inherited effects of the increased or diminished use of the different parts of the body" ([1871] 1992, 39). So while natural selection and the law of use and disuse are often represented as competing theories, it is important to note that at least some of Darwin's own description implies their status as companion theories: natural selection and the law of use and disuse support each other, because they explain how things end up going in more or less the same direction.

Comparing Lamarck and Darwin might lead us to assume that natural selection as an alternative thesis to the law of use and disuse does not make use of use. But natural selection could be understood to offer an *alternative thesis of use*. The law of use and disuse and natural selection could be described as offering different approaches to the relation of use, variation, and time. For Darwin, variations in the structure of organisms are not explained as an effect of environmental changes. Rather, variations are "spontaneous," or a matter of "chance," though both these terms are treated with caution (for a discussion, see C. Johnson 2015). So for Darwin it is not that a part comes to exist because it would be of use but that a part that happens to exist that would be of use is more likely to be preserved. Even if Lamarck and Darwin explain variation differently, both understand use as a *mode of directionality*. For Lamarck, an organism is directed by its own patterns of use that are made necessary by the environment. For Darwin, spontaneous variations in the right direction will be preserved, whereby the rightness of the direction is a matter of what *happens* to be useful in the struggle for existence.

Darwin's reflections on domestication give us more insight into his uses of use. He writes that "the key is man's power of *accumulative selection*: *nature gives* successive variations; man adds them up in certain

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directions *useful to him*. In this sense he may be said to make for himself useful breeds" ([1859] 2009, 22, emphasis mine). Darwin made use of the word *selection* in *natural selection* because of its use in animal breeding: the term that was already in use was *methodical selection*. Darwin sometimes refers to domestication as "artificial selection." Natural selection strengthens the effects of use and disuse because it preserves the parts that are more useful to an organism. In domestication or methodical selection, humans take the place of nature by doing the selecting; they breed the animals that have the characteristics that would make the animals more useful to them (and not to the animals themselves). Even if domestication is modeled on natural selection, usefulness is determined as a different kind of relation. Humans select what is useful *to them*; nature selects what is useful *to the organism*. The "to" in "useful to" is redirected away from the organism and toward another organism that is making use of it (instrumentality here could be understood as a *technique of redirection*[12]). The organism is directed in order to create "useful breeds," that is, breeds that will be useful to humans in their own struggle of existence. Human beings do not create something from nothing: rather, they have a hand in it, the "breeder's hand." They direct how organisms come to be from selecting the variations that are advantageous to them. Darwin describes, however, the artificial selection of useful breeds as an inferior activity to that of natural selection, "Natural Selection . . . is a power incessantly ready for action, and is as immeasurably superior to man's feeble efforts, as the works of Nature are to those of Art" (49). Natural selection and methodical or artificial selection both rest, however, on what Sarah Franklin has usefully called "biological plasticity" (2007, 43; 2013, 51). As she describes: "The analogy of the breeder's hand so important to Darwin's account of biological plasticity has been considerably extended in the form of animal populations bred and manipulated as models of what this plasticity can be made to do—that is, how it can be differently managed and disciplined" (2013, 51). Plasticity, the capacity of organisms to receive an influence, is instrumentalized. I return to how selection can function as a technical apparatus in the following chapters.

As we have noted for Darwin, a part does not come into existence because it would be useful. The relation of "what is of use" to "what is" is still rather and perhaps rather strangely intimate: the implication is that nothing comes to be without having been of use at some point in time. One section of chapter 6 of *The Origin of Species* is called "Utilitarian Doctrine, How Far True; Beauty, How Far Acquired." This section is

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written in response to naturalists who challenge "the utilitarian doctrine that every detail of structure has been produced for the good of its possessor" ([1859] 2009, 159). The challenges emphasize the role either of beauty or of the Creator in determining structures; Darwin cautions that both challenges would be "fatal" for his theory if they were found to be true. His defense is a utilitarian case: "Natural selection will never produce in a being any structure more injurious than beneficial to that being, for natural selection acts solely by and for the good of each" (162–63).[13] Insofar as what is useful is what is preserved, then biological structures themselves can be treated as "useful archives": what is selected was useful at one point or another in the biological life of a species such that each body carries traces of past uses. Use becomes an accumulated somatic history, a history of qualities that are acquired over time, even if what is of use now might be different from what was of use before.

The Origin of Species thus gives considerable time to the existence of useless, rudimentary, or vestigial parts, parts for which the organism has no obvious use in the present, parts that are of no obvious benefit. One of Darwin's examples was the small eyes of a blind mole: the eyes remain even though the mole no longer uses them, but they have dwindled or diminished; perhaps they are on their way to ceasing to be. The small eyes of a blind mole are an example of a vestigial part of a body that persists despite no longer being in or of use. The appendix and the tonsils are obvious human examples, although there are some who argue that these organs that have apparently become useless do have a function.[14] Vestigial organs allow us to witness what I have been describing as the strange temporalities of use; that a part can still exist even if there is no use for it implies that use can be *out of time* with a body. What is no longer of use can linger or leave traces as parts of an organism. Such parts are often referred to as *leftovers*.

I noted in the previous section how attending to use can expose queerness of time as well as biology. We can consider how a body is the site of such an exposure; a part can take time to disappear after it has ceased to be of use. Thinking about useless parts might allow us to explore the queerness of use in another way. Darwin writes:

> I presume that lowness here means that the several parts of the organisation have been but little specialised for particular functions; and as long as the same part has to perform diversified work, we can perhaps see why it should remain variable, that is, why natural selection *should not*

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> *have preserved or rejected each little deviation of form so carefully as when the part has to serve for some one special purpose*. In the same way that a knife which has to cut all sorts of things may be of almost any shape; whilst a tool for some particular purpose must be of some particular shape. Natural selection, it should never be forgotten, can act solely through and for the advantage of each being. Rudimentary parts, as it is generally admitted, are apt to be highly variable. . . . *Their variability seems to result from their uselessness*, and consequently from natural selection having had no power to check deviations in their structure. ([1859] 2009, 118–19, emphasis mine)

Darwin makes use of a used object in giving an account of the emergence of life. Objects can be used to tell stories about all manner of subjects. James made use of a lock and a key to show how repeated use or habituation leads to the overcoming of material resistance; Darwin makes use of a knife to address how more variability is enabled when something is less specialized. Knives themselves can be used to tell different stories about use; they can become blunt from use, as I noted in chapter 1. Darwin is making a different point by making use of a knife. The point is not whether it is sharp or blunt but that a knife is released from the requirement to be a particular shape when it is intended to cut "all sorts of things." If variable forms are enabled by a diversity of uses, *then specialization can be understood as the stabilization as well as restriction of form*. Indeed, returning to my discussion of Leroi-Gourhan's example of the chair that fits more bodies by fitting each body less perfectly, we could argue that advancement is understood by Darwin as the gradual loss of functional as well as biological plasticity: each part is *stabilized* by becoming more *specialized* for a particular purpose.[15] However, natural selection also depends on variability; natural selection is indifferent toward those parts that are not useful, and those parts tend to be variable. I return to the queer implications of this connection between variability and uselessness in my conclusion to this book. Suffice to say here the suggestion that uselessness enables variability makes queer a motor of history, *a freedom to deviate as a freedom from function*. To be useful is to be more subject to *checks* carried out here not by agents but by an invisible force, although one thinks of how a breeder's hand can function as a checking force.

If what would be useful changes, then an organism too must change, otherwise it would expend energy on retaining that which would not be profitable or advantageous. Darwin describes how "natural selection is

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continually trying to economize every part of the organization. If under changed conditions of life a structure, before useful, becomes less useful, its diminution will be favoured, for it will profit the individual not to have its nutriment wasted in building up an useless structure" ([1859] 2009, 117–18). This model of natural selection as an "economiser" might be how and why Marx understood Darwin to be an "apologist for industrial capitalism" despite the extent to which Marx modeled his own social history on Darwin's natural history, as Sarah Franklin has persuasively demonstrated (2013, 13). We can ask how the argument that the most useful is what is selected differs from Lucretius's example of the cup designed for usefulness's sake, mentioned in the previous chapter. Lucretius's philosophical poem demonstrated how building a theory of life from the example of designed objects leads to the mistaken fallacy that nature operates on design principles. Darwin also dismisses the idea that something is shaped for or by use, an idea that is sometimes called "finalism" as well as creationism.[16] But despite this, Darwin does in fact make use of the metaphor of the architect in describing the mechanism of natural selection. Surely this very metaphor risks the implication of nature as design? Darwin was aware of these risks (for a discussion, see C. Johnson 2015, 124–25). In the second volume of *The Variation of Animals and Plants under Domestication*, Darwin offers an interesting adoption of the architect metaphor that clarifies how he understands the relation between use and contingency. He notes:

> Throughout this chapter and elsewhere I have spoken of selection as the paramount power, yet its action absolutely depends on what we in our ignorance call spontaneous or accidental variability. Let an architect be compelled to build an edifice with uncut stones, fallen from a precipice. The shape of each fragment may be called accidental; yet the shape of each has been determined by the force of gravity, the nature of the rock, and the slope of the precipice,—events and circumstances, all of which depend on natural laws; but there is no relation between these laws and the purpose for which each fragment is used by the builder. In the same manner the variations of each creature are determined by fixed and immutable laws; but these bear no relation to the living structure which is slowly built up through the power of selection, whether this be natural or artificial selection. . . . The shape of the fragments of stone at the base of our precipice may be called accidental, but this is not strictly correct; for the shape of each depends

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on a long sequence of events, all obeying natural laws; on the nature of the rock, on the lines of deposition or cleavage, on the form of the mountain which depends on its upheaval and subsequent denudation, and lastly on the storm or earthquake which threw down the fragments. *But in regard to the use to which the fragments may be put, their shape may be strictly said to be accidental.* (1868, 248–49, emphasis mine)

We have in this quote a fascinating account of use. An architect is usually a designer and not a builder. But Darwin compels the architect to build, to make use of stones without cutting them in order to fit a design (the way a cup may be designed "for usefulness's sake" to borrow again from Lucretius's poem). The stones are thrown up, or available, according to natural laws. Determination is here a combination of events that makes things available as things to use, or even that makes some things useful for specific purposes. But use itself is described as accidental. These stones were not made in order to be used. Stones become useful to the architect *once he has begun building*: this fragment fits here, that one there; because of the shape they have, these stones happen to meet the requirements of a building that is being built. But even if they are chosen with reference to their physical properties or material qualities, which have been acquired in accordance with natural law, they were not designed to fit. *You are more likely to use a stone that happens to fit that space*. The architect is here, by analogy, natural selection: natural selection selects that which is most useful from what happens to exist. If what is selected is what would be useful, to select is to use or to put to use ("in regard to the use to which [they] may be put"). The moment of use is hap: use as happenstance; use, even, as happy. I return to the implications of Darwin's happy use of the architect metaphor in chapter 4.

I opened this chapter by nothing that the well-used expression "form follows function" came from architecture but was articulated as a law of life. The traffic between architecture and biology goes in both directions.[17] Stephen Jay Gould and Richard Lewontin introduce the concept of *spandrels* from architecture to explain biological structures. They describe spandrels as follows: "Every fan-vaulted ceiling must have a series of open spaces along the midline of the vault, where the sides of the fans intersect between the pillars. *Since the spaces must exist, they are often used for ingenious ornamental effect*" (1979, 582). They argue that spandrels are a consequence of a structural design, created because of how fans and pillars intersect. For Gould and Lewontin, spandrels demonstrate that use

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can be secondary: "a necessary by-product of fan vaulting; their appropriate use is a secondary effect" (582). They conclude, "The immediate utility of an organic structure often says nothing at all about the reason for its being" (583). In Darwin's use of the architect metaphor, even if what is selected is dependent on what would be useful given the requirements, what is built is still in accordance with what is of use; while for Gould and Lewontin, a use can be what is acquired after a structure has been built.

In this much-cited article, Gould and Lewontin are in fact calling for a return to Darwin *and* a more pluralistic account of the origin of species; in other words, a pluralistic model is understood as Darwin's legacy. As I have noted, Darwin tends to position Lamarck's law of use and disuse as a companion model to natural selection despite how he distances himself from Lamarck's use of use. This means that for Darwin, natural selection is not offered as the only mechanism to explain the origin of species. Perhaps we can think of pluralism as a *pluralism of use*. If we put these different uses of use together, they would show how use can go all the way down because use can matter at any moment of time. Use can shape what is (use as past tense); use can explain how something is selected from what is (use in the present tense); and use can be what is possible given what is (use as future tense). Use might have strange temporalities because use is how time *comes to matter*.

The Blacksmith's Arm

Most of the examples used by Darwin and Lamarck in developing their approaches to use refer to the parts of nonhuman animals. Indeed, in *Zoological Philosophy*, Lamarck's main reference to humans in exemplifying his first law of use and disuse is to diet and its effects on the size of stomachs.[18] However, a human example that is often referred to as Lamarckian is the blacksmith's strong arm. This arm has been used to illustrate both of Lamarck's laws: that what is used will be strengthened in proportion to time spent; and that the effects of repeated use will be inherited by future generations. The argument is an argument about paternity as well as generation: the sons of the blacksmith will inherit stronger arms. The strong arm becomes another well-used path, use as a direction taken that is enabled by past use, an enabling that is inherited; sons inherits the efforts of their fathers by being born with stronger arms (figure 2.4).

The blacksmith's arm is in fact so well used as an example of the effects of use that it has achieved a somewhat iconic status. Indeed, considering

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2.4. A stronger arm.

how the blacksmith's arm has been used as an example has much to teach us about the role of use in shaping fields of knowledge. Examples could be thought of as having their own "biographies of use." An example can be used as evidence *of* something. Examples can also be used as guides to help readers, by giving flesh to what might otherwise remain abstract or obscure principles. Some examples are used more than others—they are given more work to do, perhaps because they are more affective as well as effective; some examples might be remembered more easily than others.

When examples catch or become catchy, they can distract readers from the principles they are intended to exemplify. This was indeed Steven Jay Gould's concern about Lamarck's examples. Gould notes, "Every classical example—from eyeless moles, to webbed feet of water birds, to long legs of shore birds, to the blacksmith's strong right arm—ranks as a lateral deviation, not a stage on the main sequence" (2002, 188). He thus encourages readers not to overstate the significance of specific examples to Lamarck's overall argument. Gould was particularly concerned that Lamarck's example of the giraffe's long neck had become distracting. Gould even describes Lamarck's use of the giraffe's long neck as a "throwaway

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example" (305), implying that the giraffe's long neck not only has assumed too much importance but should be disregarded by careful readers. I suspect Gould describes the use of the giraffe's long neck as overuse because that example has often been used to dismiss or mischaracterize Lamarck's argument as implying voluntarism, that a part of the body can be lengthened at will or by will.

But what then did Gould have to say about the blacksmith's arm? It is here that something curious seems to have happened. In *Eight Little Piggies: Reflections in Natural History*, Gould refers to "the blacksmith's big right arm" as Lamarck's "Amanaesque metaphor" (1993, 216). When Gould discusses Aldous Huxley's *On Our Knowledge of the Causes of the Phenomena of Organic Nature*, he discusses how Huxley does not make use of the giraffe but rather the blacksmith's strong right arm as one of the two examples "emphasized by the Frenchman himself" (1999, 306). The example of the giraffe's long neck is used only once by Lamarck. However, the blacksmith's arm is not used by Lamarck at all. Lamarck does not use the example that Gould implies he emphasized. This is particularly curious given that Gould criticizes how the example of the long neck of the giraffe has been overused. His preoccupation with how the giraffe's long neck has been stretched as an example seems to have something to do with how he did not notice that Lamarck does not use the example of the blacksmith's arm at all. It is possible (given how much Gould was influenced by Darwin) that Gould was following Darwin's own emphasis. It was Darwin who used the blacksmith's arm as an example of Lamarck's law of use and disuse, which is another fascinating footnote in how use came to matter in framing the relation between Lamarck and Darwin.[19]

It is also important to note that the blacksmith's arm was widely employed throughout the nineteenth century to demonstrate the first of Lamarck's principles—the law of exercise—without reference to Lamarck. In one educational treatise, the blacksmith's arm is evoked because that is the arm that is usually evoked: "The blacksmith's arm is the standing illustration: it is said that the young lad who lifts with difficulty the sledge the first day soon finds his arm growing strong by use, and his muscles taking on the might of the steel he forges" (Yocum 1876, 545). This is an interesting instance of the principle explored of how more use can lead to more use; and by this I am referring not only to the effects of use on an arm but to *the use of the arm as an example of the effects of use*. An example is used because it has been used ("a standing illustration"). The blacksmith's arm could be considered a much-used or even overused object

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in the sense that it is being used as a figure to do a certain kind of work *whether or not the arm appears*.

Perhaps it is because the blacksmith's arm is assumed to be there that it is not noticed that the arm is missing. We could even understand Lamarck's blacksmith's arm as a phantom limb.[20] A phantom limb would typically be understood as a limb that is felt to be there after a real limb has been lost. The blacksmith's arm is not a phantom limb in this sense: the arm was not cut off. The blacksmith's arm operates as a phantom in the sense that the use of the arm covers over not only the absence of the arm but that the arm was never there. We can treat the absence of a reference to the blacksmith's arm in Lamarck as a teachable moment; we learn how *what is missing comes to matter*. The history of the use of the blacksmith's arm is a history of an impression of an arm being there. As I noted in chapter 1, the law of exercise as articulated by the psychologist Edward Thorndike suggests that an association becomes stronger through use. I also discussed how use leaves impressions with the example of the impress or indent left behind on a chair after somebody gets up.[21] The blacksmith's arm is an impression in the sense that it creates the effect of a body having been there. When an impression is shared, it becomes stronger or more distinct.

What is used most can be mostly missing. To notice that the blacksmith's arm is missing from the corpus of Lamarck's work is to learn how what is missing can still become material. In *Bodies That Matter*, Judith Butler offers a powerful exploration of the temporality of matter.[22] Butler defines matter as a "process of materialization that stabilizes over time to produce the effect of boundary, fixity, and surface we call matter" (1993, 9). Butler's approach to materialization as a process creating the effect of matter is helpful in considering the status of the blacksmith's arm. *The arm materializes as an effect of how it has been used to exemplify the effects of use*. Butler references Aristotle's discussion of matter (*materia* and *hyle*) as potentiality. She notes, "The Greek *hyle* is wood that already has been cut from trees, instrumentalized and instrumentalizable, artifactual, *on the way to being put to use*" (31–32, emphasis mine). The blacksmith's arm provides material in the sense that the arm is "on the way to being put to use." References are material; they involve a "putting to use." When the arm is constantly referred to, the references are *becoming substantial*. The references to the arm become the arm, fleshy things: a materiality created as impression, a materiality that is missing something. We could draw too upon Butler's (1990) arguments about gender as imitation to

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make sense of the use of the figure of the blacksmith's arm. Butler suggests that to perform gender is to become a copy without an original. It is not that behind the copy/reference is an original but that the copy/reference creates the impression of an original. An arm became Lamarckian without being used by Lamarck.

The arm is a phantom limb not only in the sense that it is missing from the corpus of Lamarck's work. A blacksmith's strong arm still matters or even becomes matter despite its absence because it has been exercised to do certain kinds of work. Noticing that the arm is missing allows us to ask *what the arm is doing*; the blacksmith's arm is called into existence by being called upon to work. Others before me have asked similar questions about the use of the blacksmith's arm. Archibald MacLaren in *MacMillan's Magazine* notes:

> In illustration of this, I would but point to the familiar example so frequently adduced when the value of exercise in the acquisition of strength is asserted—the Blacksmith's arm. Why is the Blacksmith's arm more powerful than that of the Clerk who uses his pen, of the Artist who handles his pencil, of the Tailor who plies his needle, or of the hand-loom Weaver who drives his shuttle? On account of the greater amount of exercise to which it is subject. Just so; but, if mere movement constituted exercise, the arms of the Clerk, Artist, Tailor, and Weaver ought also to be powerful, which we know they are not. In the wielding of their instruments their muscles have encountered comparatively no resistance, whereas the muscles of the Blacksmith's arm have had to resist the weight of a 20-lb. hammer every time they whirled it aloft; they had to arrest it in its upward flight, and to make it descend to the anvil with a speed quadruple that attending the law of gravitation. (1861, 36)

According to MacLaren, reflecting on the law of exercise, the blacksmith's arm comes to matter because of the strength required to work with the tools necessary for the task; the hammer is heavier than a pen or needle. The heaviness of the tool also relates to the hardness of the material that is being worked upon: metal rather than, say, cloth or paper. What is heavier and harder is more resistant; more force is required to work with what is more resistant to transformation. The blacksmith's arm comes to stand not only for physical or manual labor but for *how* that labor shapes or impresses upon a body. A body becomes attuned to the requirements of a job, when an arm is *becoming stronger*. It is noteworthy

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that the word *job*, suggesting "something to be done" or a "task, piece of work," is a word of uncertain origin. It is understood by some to be a variant of *gobbe*, or "mass, lump," via the sense of "a cartload." A job carries this implication forward, of heavy work. If hammering metal is a necessary part of a job, given how someone is employed, a stronger arm is *acquired as well as required*. The blacksmith's arm was thus widely used as evidence of the law of exercise, which is why it might be assumed that Lamarck used this example in making this law his first law.

When we consider how the blacksmith's strong arm is used to support Lamarck's second law about use inheritance, we can deepen our understanding of why the arm has been usefully employed. If the sons of the blacksmiths are born with stronger arms, the implication is that they have received before birth what they would need to complete the work better or more easily; the strong arm lessens what the blacksmiths' sons would need to survive or thrive within an environment. But here, through the use of this example, the environment becomes a job description: the implication being that a stronger arm would lessen the effort required by a blacksmith to do his work. The strong arm is not only how a work load is eased *but how a work load is acquired*. *For* becomes an inheritance. *For as before* is translated here into a gender and class fatalism. Fatalism can be represented as the straight line of genealogy: sons will become their fathers; they will follow in their footsteps. A description (they have strong arms) becomes a prediction (they will have strong arms) and also a command (they must have strong arms). Genealogy could be understood as a technique for straightening things out: the sons in having stronger arms are already on the way to becoming their fathers.

In receiving an expectation, the blacksmiths' sons might indeed acquire what they are assumed to possess: stronger arms. An effect becomes a cause. So it is not just the references to the blacksmith's arm that have become substantial. An arm too is required to *become substantial*. For the blacksmith's son to inherit the effects of use is *to inherit a work load*. In having stronger arms, the blacksmith's son is already equipped to become a blacksmith; his arm has a hand in deciding his future. The requirement to make use of the arm, to become a blacksmith, can thus function as a recovery mechanism: a strong arm, created as an effect of use, covers over an absence. If the blacksmith's strong arm can be understood as a phantom limb, what is received is *an idea of direction*, an idea that is all the more powerful as the body can be shaped by receiving that

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idea. If in becoming a blacksmith a strong arm is acquired, what is reproduced is an idea of inheritance.

If in this chapter I have primarily considered different *ideas of use*, I am now exploring the *uses of an idea*. Use inheritance is used to suggest that individuals receive at birth what is necessary for them to fulfill their station in life: *reception as direction*. Use inheritance is thus translated into a requirement to reproduce an inheritance. We need to ask: what else is being reproduced by such a requirement? The blacksmith's arm matters as a way of figuring not just the effects of use and disuse on organic life but the improvement of humanity. Herbert Spencer's work on use inheritance as a biological principle becomes a way of understanding how society develops as an organism. In the following quote the blacksmith's arm appears alongside other examples.

> As surely as a blacksmith's arm grows large, and the skin of a laborer's hand thick; as surely as the eye tends to become long-sighted in the sailor, and short-sighted in the student; as surely as the blind attain a more delicate sense of touch; as surely as a clerk acquires rapidity in writing and calculation; as surely as the musician learns to detect an error of a semitone amidst what seems to others a very babel of sounds; as surely as a passion grows by indulgence and diminishes when restrained; as surely as a disregarded conscience becomes inert, and one that is obeyed active; as surely as there is any efficacy in educational culture, or any meaning in such terms in habit, custom, practice; so surely must the human faculties be moulded into complete fitness for the social state; so surely must the things we call evil and immorality disappear; so surely must man become perfect. (1851, 65)

Here the blacksmith's arm appears first in a series of images, which appear as rather striking, as poses of confidence. Each person modifies their faculties by using their faculties: that the blacksmith's arm grows larger is part of the story of the perfectibility of humanity as such. The acquisition of greater skill, we must note, is directive for each subject in this list, as they exercise their faculties in accordance with their role in the social state. The eyes of the student and the eyes of the sailor are adjusted by the very nature of what they do: this is what Spencer means by "habit, custom, practice."

Use here is understood as a measure of social progress because use is strictly limited. Use increases power in accordance with function: an arm that grows stronger from being used allows the blacksmith to function

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better *as* a blacksmith. We can note as well how use is tied to best: to use one's faculties more is to become better at something, with betterment understood as a molding, as being shaped by function. Two uses of use operated alongside each other: an organ or limb is used by a person, such that it becomes stronger and better; a person becomes better at fulfilling their function because of the strengthening of an organ and limb. Intended functionality can be used to refer to uses of the body: parts of the body are used in order that a body can carry out a function *for which it is assumed it was intended*. The blacksmith in acquiring a stronger arm from the inherited effects of use becomes the arm of industry.

This adaption to function is understood by Spencer as a form of social as well as biological advancement.[23] How individuals become more adapted to their specific function within social organisations is likened to how different organs of a physical body become specialized.[24] Reginia Gagnier has usefully described Spencer's thesis as "the biologization of Progress" (2000, 98). As Spencer elaborates: "differences of function, and consequent differences of structure, at first feebly marked, slight in degree, and few in kind, become, as organization progresses, definite and numerous; and in proportion as they do this the requirements are better met" (1871, 597). Spencer makes use of Lamarck's first law to suggest that each part to be strengthened must be used, and that the strengthening of the part would enable the whole body to perform better: "Each limb, viscus, gland, or other member of an animal, is developed by exercise—by actively discharging the duties which the body at large requires of it; and similarly, any class of labourers or artisans, any manufacturing centre, or any official agency, *begins to enlarge when the community devolves on it more work*" ([1860] 1981, 413, emphasis mine). The idea that use increases capacity is used to justify workers being given "more work." The more work they are given, the more the whole body is enlarged.

If using something is understood to enable progress and refinement, then what is refined is not only an individual capacity to fulfill a function but an existing system for allocating functions. We can call this system by its name: industrial capitalism. And it is clear that when Spencer refers to "the social organism" and "the physiological division of labour" he is referring to capitalism. He describe capital as the blood that "facilitates the processes of nutrition" ([1860] 1981, 418). He also refers to large-scale manufacturing industries such as cotton, iron, and wool not only as the producers of commodities but as the sources of *nutrients* to the physical body (412–13). Indeed Spencer describes profit as the means by which

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the body is replenished and repaired: "what in commercial affairs we call *profit*, answers to the excess of nutrition over waste in the living body" (413). In other words, economic growth is understood by Spencer as biological repair. Wear and tear (to the worker's body or to the machines of industry) must be overcome in order to keep the system working.

Spencer's model of social improvement seemed to imply that all members of a society must make more use of their relevant faculties. But the task of improving humanity is made especially dependent on increasing the productivity of workers. The blacksmith's arm keeps appearing, or is given priority, as it takes up this task. In an anonymously written article from 1918, the task of improving humanity is made technical. The article begins by asking, "Did you ever watch a blacksmith at work, forging a horseshoe or some other object of wrought iron? With the tongs he draws the glowing metal from the bed of fire in the forge, and resting it upon the anvil he begins to hammer it into shape. His powerful arm turns, bends and twists to bring the hot metal into just the right position."[25] Here the "powerful arm" is in company with other objects of use: the hammer, the tongs, the forge, the anvil. These objects have a family resemblance and they work together to create another useful object: the horseshoe. The powerful arm is not the only agent in the story; it is not the only element in how things end up going in the right direction. This short description helps us deepen or thicken our account of the combination of energies and resources that enable an object to become available for use.

But the article then proceeds by suggesting that the blacksmith's arm is not powerful enough to deal with the heaviness of materials in big steel works: "A blacksmith's arm has not the strength to handle such prodigious weight." For this reason, "human ingenuity" has invented a machine. An image included with the article shows us this invention. The machine is presented as the blacksmith's arm, as extending his power and reach. The introduction of machinery to enable the increase of labor power is here narrated as the extension of the body of the laborer. The demand for him to be usefully employed is the demand for new ways for him to become the powerful arm of industry. The extraction of labor power is how use and disuse become principles of engineering.

The blacksmith's arm could be understood then not only as a phantom limb, that comes to matter, but as technology; the blacksmith's arm is a prosthetic. The blacksmith's arm becomes a living tool, teaching us how, in Sarah Franklin's (2013) terms, technology is biologized.[26] And the figure of the blacksmith's arm is used in order to tell the story of industrial

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capitalism as a story of refinement and progression—not only of bodies becoming fitter, more attuned to job requirements, but of the creation of machines that extend what bodies can do. To simplify, the blacksmith's arm was overused to tell a story of exploitation *as* a story of improvement. Challenging that story is made possible by recalling other effects of use: indeed, historical materialism made such a challenge by using use to do a different kind of work. Arlie Hochschild makes specific reference to the arm: "The factory boy's arm functioned like a piece of machinery used to produce wallpaper. His employer, regarding that arm as an instrument, claimed control over its speed and motions. In this situation, what was the relation between the boy's arm and his mind? Was his arm in any meaningful sense his *own*?" ([1983] 2003, 7; see also Zandy 2004). The laboring arm becomes not only stronger and more muscular from the requirements of work but also worn down or worn out. An arm becomes a piece of machinery to be used: use as being used; use as used up.

I noted in the previous chapter how Marx explores the history of technologies not as labor-saving devices but as systems of expenditure and exhaustion. It is living bodies that are exhausted by what they are required to do with things, with materials such as paper, metal, and cotton, in order for those things to become commodities, to create surplus value. And the materials themselves allow us to make connections between how lives are used and used up. We can think, for example, of how cotton traveled to factories in the North of England from the slave plantations in the American South. Marx notes, "As the export of cotton became of vital interest to these states, the over-working of the negro and sometimes the *using up of his life* in 7 years of labour became a factor in a calculated and calculating system" ([1867] 1990, 345, emphasis mine). The connection between slave labor and wage labor is material in many senses of material: the transformation of cotton from raw material into commodity depends on the exploitation of slave labor in the plantation as well as wage labor in the factories.

Marx is describing here the interests of the United States in *exporting* cotton. The materials that arrive at the factory to be turned into commodities are not only imported; they are already commodities that have been extracted or stolen from the lives of the enslaved. Cotton was required for the growth of the manufacturing industries and the acquisition of capitalist wealth; in other words, British manufacturing was dependent on slave labor, as Marx did recognize. Sven Beckert points out that British cotton manufacturing was "the first major industry in human

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history that lacked locally procured raw materials" (2014, 85). The history of cotton shows us how we cannot understand capitalism without reference to colonialism and slavery; the expansion of this industry depended on "slaves in the Southern United States growing cotton on land expropriated from Native Americans" (85). The necessity of slave labor and of expropriation of land from native peoples to the expansion of capitalism requires us to understand capitalism as "racial capitalism," to use Cedric J. Robinson's term. In *Black Marxism*, Robinson notes, "The British trade movement firmly seats slavery in the movement from mercantile to industrial capitalism" ([1984] 2000, 116). Marx's own descriptions of the using up of the lives of slaves need to be understood as continuous with his more frequent and familiar descriptions of the exploitation of workers in factories in the North of England. As George Shulman describes with reference to Saidiya Hartman's *Scenes of Subjection* (2002), the slave becomes "an empty vessel to be occupied, enjoyed, and used up" (2016, 35). A history of being used up is the history of racial capitalism: a history of those who are worn down, worn out; depletion *as* the extraction of surplus value or profit.

A history of the requirement to be useful is also a history of exhaustion. Use becomes for some a form of biological destiny as well as bodily depletion. What we learn from this history of use is that the idea that usefulness resides somewhere—in an arm or a body and thus also in a system—is directly related to use as being used up or overused. Race and class work together to create a set of expectations of who will be useful for what, of what some bodies are *for* or how some bodies become *for*, of who and what becomes available as resources to be used up.

The extraction of use from bodies by defining others as bodies to be used was justified as a mission to improve humanity: to become arm is to become body, to offer brawn or brute strength; *to become arm as to become biological*. A history of arms that are used, overused arms, including arms that are missing, that have been lost in service to the industrial machine, is a history of how some are used to benefit others. It is not simply that laborers are assumed to acquire stronger arms, arms that can do more because they do "do" more, but that others are *freed from that requirement*. Manual labor becomes distinguished from mental labor, a distributed distinction, a distinction that is distributed between persons and things, such that making use of mental faculties came to be understood as dependent on relative freedom from biological necessity. The blacksmith's arm was usefully employed as a means of extracting that freedom.

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The Unused Arm

The figure of the blacksmith's arm is used to tell a story of progress and advancement as the increasing refinement of the division of labor; workers become more attuned to the system by increasing their productivity and strength as workers. The figure of the blacksmith's arm also helps us understand the relation between disuse, diminishment, and degeneracy. Henry Edward Crampton's *The Doctrine of Evolution* also refers to the blacksmith's arm but then turns to the question of disuse.

> Then there are the variations of a second class, more complex in nature than the direct effects of environment,—namely, the functional results of use and disuse. A blacksmith uses his arm muscles more constantly than do most other men, and his prolonged exercise leads to an increase of his muscular capacity. All of the several organic systems are capable of considerable development by judicious exercise, as every one knows. If the functional modifications through use were unreal, then the routine of the gymnasium and the schoolroom would leave the body and the mind as they were before. Furthermore, we are all familiar with the opposite effects of disuse. Paralysis of an arm results in the cessation of its growth. When a fall has injured the muscles and nerves of a child's limb, that structure may fail to keep pace with the growth of the other parts of the body as a result of its disuse. These are simple examples of a wide range of phenomena exhibited everywhere by animals and even by the human organism, demonstrating the plasticity of the organic mechanism and its modification by functional primary factors of variation. (1912, 112–13)

Crampton makes use of the blacksmith's arm as it is usually used: as evidence of the law of exercise. But he then notes the "opposite effects" of disuse: that when an arm has paralysis, it suffers a "cessation of its growth." We can also see this coupling of inactivity or disuse and cessation in the following statement: "*Inactivity is destruction throughout the universe of things*. The human body as a whole, or as to any one part, is no exception to that boundless law. The unused arm dwindles to skin and bone; the unused lungs soon weaken, then rot away" (Hall 1869, 224). When something stops being used, the implication is that it ceases to be. This is how use becomes an expression of a commitment to life.

The figure of the unused arm is rather striking. *Willful Subjects* (2014) began with another striking arm that appears in a Grimm story. In the

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story, a child disobeys her mother and is allowed to die. Her death is punishment for her willfulness. But her arm keeps coming up from the grave until it too is beaten down. The arm in the Grimm story inherits willfulness from a child; it keeps coming up despite the death of the body of which it is a part. Once I caught sight of the willful arm, arms had my attention; the book itself became a call to arms. And perhaps it was because arms had my attention that I was rather struck by the use of the blacksmith's arm and that I noticed that the arm was missing. Even if the willful arm and the unused arm appear very differently—one full of life and vitality, the other emptied of life and dwindling—they both can have deadly consequences. Arms that are too active, too alive, and arms that are too inactive, not alive enough, are called upon; they work as death threats, signaling that the consequence of not obeying the instruction to work would be deadly.

It is important then to think about how unused arms are used, what they are being asked to do.[27] Scott Nearing and Nellie M. S. Nearing's *Woman and Social Progress*, published in 1912, makes use of the figure of the unused arm.

> It is true of any animal that disuse means decay; the arm, held rigid for a year, would prove an indifferent member; this is equally true of any other faculty. Disuse involves decay,—physical, mental, spiritual.
>
> The powers of the will,—the positive forces of the individual which make up character,—are no exception to this rule. The man who idles becomes an idler; the women who make no effort become colorless. Like the unused arm, they degenerate through lack of functioning. (1912, 145)

The arm stands out or stands up in becoming rigid. The arm becomes an indifferent member of the body when it is not used; the failure to contribute to the whole body becomes the loss of the capacity to make a contribution. Not to function would be to cease to be functional. In exercising the figure of the unused arm, the text evokes both the promise of utility and the risk of degeneration. *What would stop an organ or a body from working is not working*. What would make an organ or body work is work. I return to how employment becomes a duty in the next chapter. We might note as well how the will becomes what must work to generate the character of an individual: if the unused arm is an indifferent member of an individual body, an unwilling individual would become an indifferent member of the social body.

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So disuse becomes degeneracy when something is not being used that is *supposed* to be used. It is this "supposed to" that renders useful a morally weighted term even when it appears to be neutral or at least just a matter of what happens, like the stone that happens to be useful because it fits the space in a wall. We are back to how the blacksmith's arm appears in the form of an assumption and is given the form of an assumption. The strength of the arm becomes a measuring stick for the improvement of humanity because this arm is supposed to be laboring for that improvement. Disuse becomes degeneracy *when something that is assumed to be for is not being used for what it is for*. I noted in chapter 1 how an unused path reveals a paradox: for something to be usable, it must be used. An arm becomes the path for the laborer; you must follow the right path by keeping the arm in use, by strengthening the arm. The unused arm becomes a symbol of waste but also of death: the laborers who do not make use of their arms, who do not lend their hands to the master, cannot be supported without compromising the health of the social body.

Eugenics made such a profound use of use given how disuse was associated with death. Eugenics was first coined as a term by Francis Galton in 1883, deriving from "well born." Galton was Charles Darwin's cousin. One suspects this kinship story matters. Galton presented a paper on eugenics to the newly formed Sociological Society on May 16, 1904; the lecture was chaired by his student Karl Pearson, who became the first chair of eugenics at London University. This is another kinship story, a story of inheritance. I return to the significance of how eugenics was housed and inherited by London University in chapter 4. Both Galton and Pearson were committed to concepts of hard inheritance; indeed, Pearson was explicitly anti-Lamarckian, arguing against the inheritance of acquired characteristics. In coining the term *eugenics*, Galton was advocating for the improvement of humanity. He wrote, "The aim of eugenics is to bring as many influences as can be reasonably employed, to cause the useful classes in the community to contribute *more* than their proportion to the next generation" (1904, 1). Eugenics became a system for ensuring that the more useful are reproduced more.

It is not accidental (and I will return to the question of what is at stake in my use of accidental) that Spencer's commitment to utility as progress also led him in a eugenicist direction. He wrote, "Besides an habitual neglect of the fact that the quality of a society is physically lowered by the artificial preservation of its feeblest members, there is a habitual neglect

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of the fact that the quality of a society is lowered morally and intellectually, by the artificial preservation of those who are least able to take care of themselves" (1873, 606). Indeed, Spencer suggests that unworthiness ought to be deadly: "For if the unworthy are helped to increase, by shielding them from that mortality which their unworthiness would naturally entail, the effect is to produce, generation after generation, a greater unworthiness" (606). Spencer suggests that disuse would or should gradually lead to diminishment and death: "From diminished use of self-conserving faculties, already deficient, there must result, in posterity, the smaller amounts of self-conserving faculties" (606). Certain classes of people are assumed as unable or unwilling to use their faculties, and as such, Spencer argues, by the law of use and disuse, *they should cease to be*. The link made between disuse and unworthiness is fatal. In following these links, we are accounting for fatalities, even when we challenge how they were understood to be caused.

To track how eugenics made use of use would enable us to explore not how eugenics was an extreme position that can be held apart from wider social thought but how eugenicist ideas were widely diffused and disseminated: another history of use. Eugenicist ideas were *in use* in both senses I described in the previous chapter; they had currency and were in circulation, including within writing that might be understood as socially progressive. For example, Scott and Nellie M. S. Nearing's call for the education of women not only made use of use but also used the language of eugenics. They argued that middle-class and white women should make use of their faculties (that these women should not be allowed to dwindle like unused arms) so they could improve the race through making better reproductive choices. The call to make use of use was also about the dangers of uselessness spreading.

We can return here to how the laws of use and disuse were used to exemplify the success of social reproduction: the workers by working, by using their arms, shape the bodies of future workers. The danger in eugenicist writing is typically framed through reproduction—that useless parts would be too reproductive, thus spreading their uselessness throughout the whole social body. This reproductive success is explained as brought about by artificial means: the useless have been stopped from disappearing by the use of medicine and welfare. In Darwin's later writings (probably as a result of the influence of Galton—whom he cites—on his own work), this eugenicist impulse is also explicit. Darwin writes:

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> With savages, the weak in body or mind are soon eliminated; and those that survive commonly exhibit a vigorous state of health. We civilised men, on the other hand, do our utmost to check the process of elimination; we build asylums for the imbecile, the maimed, and the sick; we institute poor-laws; and our medical men exert their utmost skill to save the life of every one to the last moment. There is reason to believe that vaccination has preserved thousands, who from a weak constitution would formerly have succumbed to small-pox. Thus the weak members of civilised societies propagate their kind. No one who has attended to the breeding of domestic animals will doubt that this must be highly injurious to the race of man. It is surprising how soon a want of care, or care wrongly directed, leads to the degeneration of a domestic race; but excepting in the case of man itself, hardly any one is so ignorant as to allow his worst animals to breed. ([1871] 1992, 161)

It is still important today to challenge such an account of the history of poor laws. I will be referencing the inhumanity and cruelty of the poor laws in the following chapter; the very techniques that are represented here as checking "the process of elimination" led to elimination or the extraction of utility from death. Although Darwin does not argue for the elimination of the weak, the conclusions are stark: the reproduction of the weak (with weakness being defined, at least here, as constitutional) would weaken the species.

In order not to degenerate, a hand is called forth: the eugenicist hand as the breeder's hand; the strong must reproduce more. H. G. Wells's *Anticipations of the Reaction of Mechanical and Scientific Progress upon Human Life and Thought* also makes such a call to hands. This text frequently refers to the idea of the "social body," and how much the improvement of humanity requires the improvement of all its parts. It also frequently refers to "the abyss," a mass of unemployed people who are deemed useless and not worthy of being reproduced. Uselessness can be a deadly assignment. For Wells argues that the abyss can infect the whole body: "It has become apparent that whole masses of human population are, as a whole, inferior in their claim upon the future" and that their "characteristic weaknesses are contagious and detrimental in the civilizing fabric, and that their range of incapacity tempts and demoralises the strong" (1901, 314). Wells concludes, "To give them equality is to sink to their level, to protect and cherish them is to be swamped in their fecundity." The strong, according to Wells, need to select the strong by avoiding a

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politics of charity and equality that would artificially enable the weak to flourish. It is not just that what is selected is what is useful but that selection becomes a moral obligation of those already valued as strong: the "new republic" will be shaped by the procreation of "what is fine and efficient and beautiful in humanity" (314). The new ethics will "let die" a "multitude of contemptible and silly creatures, fear-driven and helpless and useless, unhappy or hatefully happy in the midst of squalid dishonour, feeble, ugly, inefficient, born of unrestrained lusts, and increasing and multiplying through sheer incontinence and stupidity" (323). Wells calls explicitly for the "euthanasia of the weak and sensual"(332).[28]

Use and uselessness (and their cousins efficiency and inefficiency) appear here as one value system among others such as happiness and unhappiness, beauty and ugliness, reason and sensuality. What matters is how they become attached to each other. I noted in *The Promise of Happiness* (2010) how the wretched have a history—how the history of happiness needs to be rewritten from the point of view of the wretch, of those deemed unhappy or, to use Wells's morbid terms, if happy, hatefully so. The language of utility, even if less binding or affective than happiness, is directive. Utility might be all the more directive for appearing all the less affective.

We can note that usefulness, unlike happiness, is not defined as a quality of the elite, those who have a power to enforce an assumption of being "the best part." Those who are designated useless are (mostly) those who are required to be useful, the laboring population. We have behind us strong critiques of the judgment of uselessness. We could take as an example Simone de Beauvoir's play *The Useless Mouths* ([1945] 2011). In their introduction to the play, Liz Stanley and Catherine Naji note, "The play deals with the ethical consequences of treating some persons as worthless and useless" (2011, 11). One thinks back to Pascal's useless foot that had forgotten it was part of the body. Here mouths are useless because of what they require from a body, to be nourished. There is a siege and people are starving. Beauvoir shows how ethical indifference toward, and violence against, those who need to be fed to survive is enabled by understanding that need as a deprivation of what should belong to others (to deprive others of something that is theirs to use). Tactics for demonizing others as useless turn violence into a moral right until a death of a population can be willed as the common good.

Useful is what some must become in order to avoid the consequences of being judged useless. In the following chapter, I consider how usefulness

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thus becomes a moral duty that required the full employment of the worker's limbs. We need to remember how the language of use, in biology as well as psychology, is the language of strength: to use something is understood as to make it stronger. The judgment of uselessness is brought to a eugenic conclusion, a conclusion that those who are weaker should be eliminated, in one way or another, by positive or negative means, a conclusion that was not necessary or inevitable but was not entirely accidental. Or perhaps we might say, following Darwin, that if its use was accidental, it still followed laws of determination in having become available to be used for this purpose.

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